MATH 231 – Linear Algebra I

The Matrix Unit, Part 6 ▪ The Null Space and Rank–Nullity

Kiely Hall 326 ▪ Instructor: Kevin Bedoya

About this document. This is Part 6 of the matrix unit, Sections 30–34. Section numbers run continuously through the unit (Part 2's opening table lists them all). References into the vector unit always name it.

Assumed. The column space, row space and rank (§10–§12); a matrix as a linear transformation (§13); the 2×2 determinant and its link to rank (§14, §22); projections and invariant lines met in Part 5. From the vector unit: linear independence, basis and dimension (§16–§18, §24), and subspaces (§19).

What you should be able to do afterwards. Say whether a matrix has full rank, full column rank, or full row rank; recognise invariant subspaces of simple matrices; find the null space of a small matrix and its nullity; describe the full solution set of $\mathbf{Ax} = \mathbf{b}$ as one solution plus the null space; state and prove the rank–nullity theorem and use it to predict nullities; prove that $\operatorname{rank}(\mathbf{A}^\mathsf{T}\mathbf{A}) = \operatorname{rank}(\mathbf{A})$ and that row rank equals column rank; and invert a 2×2 matrix, or say why it cannot be inverted.

30. Column and row space, revisited

30.1. Full rank and rank deficiency

§12.2 showed that $\operatorname{rank}(\mathbf{A}) \leq \min(m, n)$ for $\mathbf{A} \in \mathbb{R}^{m \times n}$. Reaching that bound is the best a matrix can do, and it has a name.

Definition 30.1 (Full rank)**.** $\mathbf{A} \in \mathbb{R}^{m \times n}$ has **full rank** if $\operatorname{rank}(\mathbf{A}) = \min(m, n)$, and is **rank-deficient** otherwise. More specifically, $\mathbf{A}$ has **full column rank** if $\operatorname{rank}(\mathbf{A}) = n$ (its columns are independent), and **full row rank** if $\operatorname{rank}(\mathbf{A}) = m$ (its rows are independent, or equivalently $\operatorname{col}[\mathbf{A}] = \mathbb{R}^m$).

For a tall matrix ($m > n$), full rank means full column rank. For a wide one ($m < n$), it means full row rank. For a square one, it means both.

Example 30.1.

- $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$: rank 2, full rank.
- $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$: rank $1 < 2$, rank-deficient.
- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$: rank 2, full column rank, not full row rank (its column space is a plane in $\mathbb{R}^3$).
- $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$: rank 2, full row rank.
- $\mathbf{uv}^\mathsf{T}$ with $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $n \geq 2$: rank 1, rank-deficient.

30.2. From one space into the other

A matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ takes inputs in $\mathbb{R}^n$ and produces outputs in $\mathbb{R}^m$ (§13.1). Its transpose $\mathbf{A}^\mathsf{T} \in \mathbb{R}^{n\times m}$ goes the other way, from $\mathbb{R}^m$ back to $\mathbb{R}^n$. Each space has two of the subspaces met so far:

Space	What lives there	What lands there
$\mathbb{R}^n$ (inputs of $\mathbf{A}$)	the rows of $\mathbf{A}$, and row$[\mathbf{A}]$	outputs of $\mathbf{A}^\mathsf{T}$: col$[\mathbf{A}^\mathsf{T}]$ = row$[\mathbf{A}]$
$\mathbb{R}^m$ (outputs of $\mathbf{A}$)	the columns of $\mathbf{A}$, and col$[\mathbf{A}]$	outputs of $\mathbf{A}$: col$[\mathbf{A}]$

The transpose does not undo the matrix. In general $\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} \neq \mathbf{x}$, and an orthogonal matrix is the exception (§20). But $\mathbf{A}^\mathsf{T}$ always carries $\mathbb{R}^m$ into the row space, the subspace of the input side that $\mathbf{A}$ actually "reads" (§11.3).

30.3. Images of subspaces

A matrix sends whole subspaces to subspaces. If $W \subseteq \mathbb{R}^n$ is a subspace, its **image**

$$\mathbf{A}(W) = \{\, \mathbf{A}\mathbf{w} : \mathbf{w} \in W \,\}$$

is a subspace of $\mathbb{R}^m$, by the same three lines that proved §10.2. If $W = \text{span}\{\mathbf{w}_1, \ldots, \mathbf{w}_k\}$ then $\mathbf{A}(W) = \text{span}\{\mathbf{A}\mathbf{w}_1, \ldots, \mathbf{A}\mathbf{w}_k\}$, by linearity. So the image is spanned by k vectors, and its dimension is at most $\dim W$: a matrix can lower dimension but never raise it. A line goes to a line or to $\{\mathbf{0}\}$; a plane goes to a plane, a line, or $\{\mathbf{0}\}$. The column space is the image of the whole input space, col$[\mathbf{A}] = \mathbf{A}(\mathbb{R}^n)$.

30.4. Invariant subspaces

For a square matrix the inputs and outputs live in the same space, so it makes sense to ask whether a subspace is sent into *itself*.

Definition 30.2 (Invariant subspace)**.** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$. A subspace $W \subseteq \mathbb{R}^n$ is **invariant** under $\mathbf{A}$ if $\mathbf{A}\mathbf{w} \in W$ for every $\mathbf{w} \in W$.

Three are invariant under every square matrix: $\{\mathbf{0}\}$, all of $\mathbb{R}^n$, and col$[\mathbf{A}]$ (whatever goes in, $\mathbf{A}$ of it is in the column space). The interesting ones depend on the matrix.

Example 30.2.

(a) The projection $\mathbf{P}_\mathbf{u}$ of §26: span$\{\mathbf{u}\}$ is invariant ($\mathbf{P}_\mathbf{u}\mathbf{u} = \mathbf{u}$), and so is the perpendicular line, which is sent to $\mathbf{0}$, a vector of every subspace.

(b) A diagonal matrix diag(s_1, s_2): each coordinate axis is invariant, since each is stretched along itself (§16.3).

(c) The shear $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$, $k \neq 0$: the x-axis is invariant ($\mathbf{e}_1 \mapsto \mathbf{e}_1$); the y-axis is not ($\mathbf{e}_2 \mapsto [\,k, 1\,]$).

(d) The rotation $\mathbf{R}_z(\theta)$ of $\mathbb{R}^3$ (§25.1): the z-axis and the xy-plane are both invariant.

(e) The rotation $\mathbf{R}_\theta$ of $\mathbb{R}^2$, with θ not a multiple of π: *no* line through the origin is invariant, since every nonzero vector is turned off its own line. The only invariant subspaces are $\{\mathbf{0}\}$ and $\mathbb{R}^2$.

(f) A reflection $2\mathbf{P_u} - \mathbf{I}$ (§28.4): the mirror span$\{\mathbf{u}\}$ and the line perpendicular to it.

An invariant *line* span$\{\mathbf{v}\}$ is special. For it to be invariant, $\mathbf{Av}$ must lie on the line, which means $\mathbf{Av} = \lambda\mathbf{v}$ for some number λ: the matrix only stretches $\mathbf{v}$. A nonzero vector with this property is an *eigenvector*, and the number λ an *eigenvalue*. They are the subject of the next unit, and (e) above is the first sign that some real matrices have none.

31. The null space

31.1. The definition

§14.5 found a whole line of the plane that $\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$ crushes to $\mathbf{0}$. Every matrix has such a set.

Definition 31.1 (Null space, nullity)**.** The **null space** of $\mathbf{A} \in \mathbb{R}^{m\times n}$ is the set of inputs that $\mathbf{A}$ sends to zero:

$$\text{null}[\mathbf{A}] = \{\, \mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{0} \,\}.$$

It is also called the **kernel** of $\mathbf{A}$; other books write $N(\mathbf{A})$ or $\ker \mathbf{A}$. Its dimension is the **nullity**, $\text{nullity}(\mathbf{A}) = \dim \text{null}[\mathbf{A}]$.

The equation $\mathbf{Ax} = \mathbf{0}$ is a **homogeneous** system: its right-hand side is zero. It always has at least the solution $\mathbf{x} = \mathbf{0}$. Part 8 solves such systems systematically; here the examples are small enough to solve by hand.

31.2. It is a subspace

Theorem 31.1. null$[\mathbf{A}]$ *is a subspace of* $\mathbb{R}^n$.

Proof. $\mathbf{A0} = \mathbf{0}$. If $\mathbf{Ax} = \mathbf{0}$ and $\mathbf{Ay} = \mathbf{0}$, then $\mathbf{A}(\mathbf{x}+\mathbf{y}) = \mathbf{Ax}+\mathbf{Ay} = \mathbf{0}$, and $\mathbf{A}(c\mathbf{x}) = c\,\mathbf{Ax} = \mathbf{0}$. □

A common confusion (Which space?). The null space lives among the *inputs*, in $\mathbb{R}^n$. The column space lives among the *outputs*, in $\mathbb{R}^m$. For a non-square matrix they are in different spaces and cannot even be compared. For a square matrix they can, and they are usually different subspaces.

31.3. Simple examples

Example 31.1.

(a) $\mathbf{I}$: $\mathbf{Ix} = \mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$. null$[\mathbf{I}] = \{\mathbf{0}\}$, nullity 0.

(b) The $m \times n$ zero matrix: every $\mathbf{x}$ is sent to $\mathbf{0}$. The null space is $\mathbb{R}^n$, nullity n.

(c) $\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$: both rows say $x_1 + 2x_2 = 0$, so $x_1 = -2x_2$ and the null space is the line span$\{[\,-2, 1\,]\}$, nullity 1.

(d) The projection $\mathbf{P_u}$ in $\mathbb{R}^2$: by §26.4 it crushes exactly the vectors perpendicular to $\mathbf{u}$, so null$[\mathbf{P_u}] = \text{span}\{\mathbf{v}\}$ with $\mathbf{v} \perp \mathbf{u}$, nullity 1.

(e) $\mathbf{A} = \begin{bmatrix}1 & 0 & 2\\ 0 & 1 & 3\end{bmatrix}$: the equations are $x_1 + 2x_3 = 0$ and $x_2 + 3x_3 = 0$. Let $x_3 = t$, any number; then $x_1 = -2t$ and $x_2 = -3t$. The null space is the line span$\{[\,-2, -3, 1\,]\}$ in $\mathbb{R}^3$, nullity 1. Check: $\mathbf{A}[\,-2, -3, 1\,] = [\,-2 + 2,\ -3 + 3\,] = \mathbf{0}$.

(f) An outer product $\mathbf{u}\mathbf{v}^{\mathsf{T}}$ with $\mathbf{u} \neq \mathbf{0}$: $(\mathbf{u}\mathbf{v}^{\mathsf{T}})\mathbf{x} = \mathbf{u}\,(\mathbf{v}^{\mathsf{T}}\mathbf{x})$, which is $\mathbf{0}$ exactly when $\mathbf{v}^{\mathsf{T}}\mathbf{x} = 0$. The null space is everything perpendicular to $\mathbf{v}$. In $\mathbb{R}^3$ that is a plane, nullity 2.

(g) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$: $\det = -2 \neq 0$, the columns are independent, and (next subsection) the null space is $\{\mathbf{0}\}$.

31.4. Null space and independence

By the column reading, $\mathbf{A}\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$. So a nonzero vector in the null space is exactly a nontrivial combination of the columns that equals $\mathbf{0}$, which is a dependence among the columns (the vector unit, §18.4).

Theorem 31.2. $\text{null}[\mathbf{A}] = \{\mathbf{0}\}$ *if and only if the columns of* $\mathbf{A}$ *are linearly independent, i.e.* $\mathbf{A}$ *has full column rank.*

And a vector in the null space *records* a dependence. In Example 31.1(e), $[\,-2, -3, 1\,]$ says $-2\mathbf{a}_1 - 3\mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}$, that is, $\mathbf{a}_3 = 2\mathbf{a}_1 + 3\mathbf{a}_2$.

31.5. What the matrix forgets

Two inputs give the same output exactly when they differ by a null-space vector:

$$\mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{y} \iff \mathbf{A}(\mathbf{x} - \mathbf{y}) = \mathbf{0} \iff \mathbf{x} - \mathbf{y} \in \text{null}[\mathbf{A}].$$

The null space measures exactly what the matrix forgets. If it is $\{\mathbf{0}\}$, different inputs always give different outputs, and the matrix is called **one-to-one**. If it is larger, whole families of inputs are merged into one output, and the information that told them apart is gone.

31.6. The shape of a solution set

Theorem 31.3. *Suppose* $\mathbf{x}_p$ *is one solution of* $\mathbf{A}\mathbf{x} = \mathbf{b}$*. Then the solutions are exactly the vectors*

$$\mathbf{x} = \mathbf{x}_p + \mathbf{z}, \qquad \mathbf{z} \in \text{null}[\mathbf{A}].$$

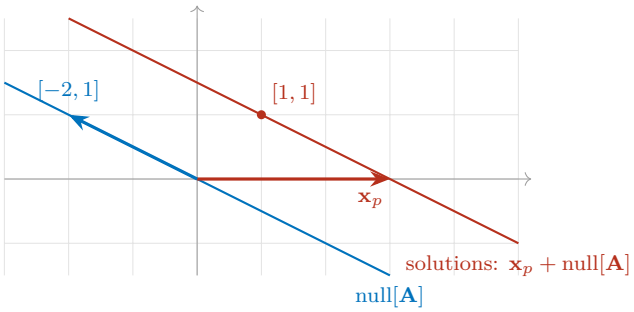
Proof. If $\mathbf{z} \in \text{null}[\mathbf{A}]$ then $\mathbf{A}(\mathbf{x}_p + \mathbf{z}) = \mathbf{b} + \mathbf{0} = \mathbf{b}$. Conversely, if $\mathbf{A}\mathbf{x} = \mathbf{b}$ then $\mathbf{A}(\mathbf{x} - \mathbf{x}_p) = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so $\mathbf{z} = \mathbf{x} - \mathbf{x}_p$ is in the null space. □

This completes the sorting that §10.3 began. For any system:

- **no solution** if $\mathbf{b} \notin \text{col}[\mathbf{A}]$;
- **exactly one** if $\mathbf{b} \in \text{col}[\mathbf{A}]$ and $\text{null}[\mathbf{A}] = \{\mathbf{0}\}$;
- **infinitely many** if $\mathbf{b} \in \text{col}[\mathbf{A}]$ and $\text{null}[\mathbf{A}] \neq \{\mathbf{0}\}$, since a nonzero null space contains a whole line.

These are the three outcomes of the vector unit's review, explained now for systems of every size.

Example 31.2. $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}\mathbf{x} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ has the solution $\mathbf{x}_p = [\,3, 0\,]$. The null space is $\text{span}\{[\,-2, 1\,]\}$, so every solution is $[\,3, 0\,] + t[\,-2, 1\,]$. In the picture the solutions form a line parallel to the null space, shifted off the origin by $\mathbf{x}_p$. Any other solution, such as $[\,1, 1\,]$ ($t = 1$), would do as $\mathbf{x}_p$.



31.7. The null space is orthogonal to the row space

By the row reading (§2.3, §11.3), $\mathbf{Ax} = \mathbf{0}$ says that every row of $\mathbf{A}$ has dot product 0 with $\mathbf{x}$. A vector orthogonal to each of the rows is orthogonal to every combination of them, by linearity of the dot product (the vector unit, §21.1). So:

Theorem 31.4. *Every vector of* null[$\mathbf{A}$] *is orthogonal to every vector of* row[$\mathbf{A}$].

In Example 31.1(e), $[-2, -3, 1]$ is orthogonal to both rows: $-2+0+2=0$ and $0-3+3=0$. The null space and the row space are two perpendicular subspaces of the input space $\mathbb{R}^n$.

32. The rank–nullity theorem

32.1. A pattern

Collect the examples so far.

Matrix	n (columns)	rank	nullity
$\mathbf{I}_2$	2	2	0
2×2 zero matrix	2	0	2
$\left[\begin{smallmatrix}1&2\\2&4\end{smallmatrix}\right]$	2	1	1
$\mathbf{P_u}$ in $\mathbb{R}^2$	2	1	1
$\left[\begin{smallmatrix}1&0&2\\0&1&3\end{smallmatrix}\right]$	3	2	1
$\mathbf{uv}^\mathsf{T}$ in $\mathbb{R}^{3\times3}$	3	1	2

In every row, rank plus nullity is the number of columns. The intuition is a budget. The input space has n dimensions. The matrix crushes some of them to $\mathbf{0}$, the nullity, and the rest survive into the output as the column space.

Theorem 32.1 (Rank–nullity)**.** *For every* $\mathbf{A} \in \mathbb{R}^{m\times n}$,

$$\boxed{\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n,}$$

the number of columns of $\mathbf{A}$.

32.2. What the proof uses

From the vector unit. Three facts about dimension. The vector unit proved them in $\mathbb{R}^2$ and $\mathbb{R}^3$ and used them in $\mathbb{R}^n$ (§17, §24), and they are taken as given here, as Part 1 §1.1 took the fact that n independent vectors span $\mathbb{R}^n$.

F1. Every subspace of $\mathbb{R}^n$ has a basis.

F2. Any linearly independent set of vectors in a subspace can be extended (by adding vectors) to a basis of that subspace.

F3. All bases of a subspace have the same number of vectors, its dimension.

32.3. The proof

Proof of Theorem 32.1. Let $p = \text{nullity}(\mathbf{A})$ and let $\mathbf{k}_1, \dots, \mathbf{k}_p$ be a basis of null[$\mathbf{A}$] (F1). These are independent vectors of $\mathbb{R}^n$, so by F2 they can be extended to a basis of $\mathbb{R}^n$,

$$\mathbf{k}_1, \dots, \mathbf{k}_p,\ \mathbf{w}_1, \dots, \mathbf{w}_r, \qquad p + r = n \text{ by F3.}$$

The claim is that $\mathbf{A}\mathbf{w}_1, \dots, \mathbf{A}\mathbf{w}_r$ is a basis of col[$\mathbf{A}$]. Then $\text{rank}(\mathbf{A}) = r = n - p$, which is the theorem.

They span col[$\mathbf{A}$]. Any vector of the column space is $\mathbf{A}\mathbf{x}$ for some $\mathbf{x}$. Write $\mathbf{x}$ in the basis, $\mathbf{x} = c_1\mathbf{k}_1 + \cdots + c_p\mathbf{k}_p + d_1\mathbf{w}_1 + \cdots + d_r\mathbf{w}_r$. Since $\mathbf{A}\mathbf{k}_i = \mathbf{0}$,

$$\mathbf{A}\mathbf{x} = d_1\mathbf{A}\mathbf{w}_1 + \cdots + d_r\mathbf{A}\mathbf{w}_r.$$

They are independent. Suppose $d_1\mathbf{A}\mathbf{w}_1 + \cdots + d_r\mathbf{A}\mathbf{w}_r = \mathbf{0}$. Then $\mathbf{A}(d_1\mathbf{w}_1 + \cdots + d_r\mathbf{w}_r) = \mathbf{0}$, so $d_1\mathbf{w}_1 + \cdots + d_r\mathbf{w}_r$ is in the null space and is a combination of the $\mathbf{k}$'s:

$$d_1\mathbf{w}_1 + \cdots + d_r\mathbf{w}_r = c_1\mathbf{k}_1 + \cdots + c_p\mathbf{k}_p.$$

Moving everything to one side gives a combination of the basis vectors $\mathbf{k}_1, \dots, \mathbf{k}_p, \mathbf{w}_1, \dots, \mathbf{w}_r$ equal to $\mathbf{0}$. A basis is independent, so every coefficient is zero, in particular every d_j. □

The proof also says *which* part of the input survives: the directions $\mathbf{w}_j$ added to the null space are carried one-for-one onto a basis of the column space.

32.4. Many examples

The theorem predicts nullities without solving anything, and it predicts ranks from null spaces found by inspection.

Example 32.1.

(a) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ has rank 2 (§12.3), so nullity $3 - 2 = 1$. The dependence $\mathbf{a}_1 + \mathbf{a}_2 - \mathbf{a}_3 = \mathbf{0}$ gives the null space: span$\{[\,1, 1, -1\,]\}$.

(b) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$: rank 2, so nullity $2 - 2 = 0$. Independent columns, trivial null space.

(c) The $1 \times n$ row $\begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}$: rank 1, so nullity $n - 1$. Its null space is the set of vectors whose entries sum to 0, a "plane" of dimension $n - 1$ in $\mathbb{R}^n$.

(d) $\mathbf{R}_\theta$: rank 2, nullity 0. A rotation crushes nothing.

(e) $\mathbf{P}_\mathbf{u}$ in $\mathbb{R}^n$: rank 1, so nullity $n - 1$. The null space is everything perpendicular to $\mathbf{u}$.

(f) $\mathbf{I} - \mathbf{P_u}$ in $\mathbb{R}^n$: it kills exactly the multiples of $\mathbf{u}$, so nullity 1 and rank $n-1$.

(g) The 3×3 shear $\begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$: independent columns, rank 3, nullity 0.

(h) $\mathbf{u}\mathbf{v}^\mathsf{T} \in \mathbb{R}^{m \times n}$ with $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$: rank 1, nullity $n-1$.

(i) *Any* 4×6 matrix has rank at most 4, so nullity at least $6 - 4 = 2$. Without seeing a single entry, we know $\mathbf{A}\mathbf{x} = \mathbf{0}$ has a plane's worth of solutions at least.

32.5. Consequences

Corollary 32.2 (Square matrices)**.** *For $\mathbf{A} \in \mathbb{R}^{n \times n}$ the following are equivalent:* $\text{null}[\mathbf{A}] = \{\mathbf{0}\}$*;* $\text{rank}(\mathbf{A}) = n$*;* $\text{col}[\mathbf{A}] = \mathbb{R}^n$*. In words, for a square matrix, columns that are independent also span, and columns that span are also independent.*

Proof. Rank–nullity gives the first equivalence. For the second, an n-dimensional subspace of $\mathbb{R}^n$ is all of $\mathbb{R}^n$: by F2 its basis extends to a basis of $\mathbb{R}^n$, but by F3 that basis already has n vectors, so nothing is added. □

This is the fact Part 1 §1.1 took on trust: n independent vectors in $\mathbb{R}^n$ form a basis. It follows from F1–F3 and rank–nullity. For a square system $\mathbf{A}\mathbf{x} = \mathbf{b}$ it means: a unique solution for one $\mathbf{b}$, a unique solution for every $\mathbf{b}$, and independent columns all stand or fall together.

Corollary 32.3 (Wide and tall matrices)**.** *Let $\mathbf{A} \in \mathbb{R}^{m \times n}$.*

- *If $m < n$ (more unknowns than equations),* $\text{nullity}(\mathbf{A}) \geq n - m > 0$*: the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$ has nonzero solutions.*
- *If $m > n$ (more equations than unknowns),* $\text{rank}(\mathbf{A}) \leq n < m$*, so* $\text{col}[\mathbf{A}] \neq \mathbb{R}^m$*: some right-hand sides $\mathbf{b}$ have no solution.*

33. A corollary: $\text{rank}(\mathbf{A}^\mathsf{T}\mathbf{A}) = \text{rank}(\mathbf{A})$

33.1. The same null space

For any $\mathbf{A} \in \mathbb{R}^{m \times n}$, the product $\mathbf{A}^\mathsf{T}\mathbf{A}$ is square, $n \times n$, and symmetric (§8.6, P4). It turns up wherever lengths are squared, as in §20 and Part 9. Its null space is the null space of $\mathbf{A}$.

Theorem 33.1. $\text{null}[\mathbf{A}^\mathsf{T}\mathbf{A}] = \text{null}[\mathbf{A}]$.

Proof. If $\mathbf{A}\mathbf{x} = \mathbf{0}$, then $\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} = \mathbf{A}^\mathsf{T}\mathbf{0} = \mathbf{0}$. So $\text{null}[\mathbf{A}] \subseteq \text{null}[\mathbf{A}^\mathsf{T}\mathbf{A}]$.

Conversely, suppose $\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} = \mathbf{0}$. Multiply on the left by $\mathbf{x}^\mathsf{T}$:

$$0 = \mathbf{x}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} = (\mathbf{A}\mathbf{x})^\mathsf{T}(\mathbf{A}\mathbf{x}) = \|\mathbf{A}\mathbf{x}\|^2,$$

using M6 and §7.3. A vector of length zero is the zero vector, so $\mathbf{A}\mathbf{x} = \mathbf{0}$. □

33.2. The same rank

Corollary 33.2. *For every* $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\operatorname{rank}(\mathbf{A}^\mathsf{T}\mathbf{A}) = \operatorname{rank}(\mathbf{A})$.

Proof. $\mathbf{A}$ and $\mathbf{A}^\mathsf{T}\mathbf{A}$ both have n columns, and by Theorem 33.1 the same nullity. By rank–nullity, $\operatorname{rank}(\mathbf{A}^\mathsf{T}\mathbf{A}) = n - \operatorname{nullity}(\mathbf{A}^\mathsf{T}\mathbf{A}) = n - \operatorname{nullity}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$. □

Example 33.1. $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$ has rank 1 (the second column is twice the first), and $\mathbf{A}^\mathsf{T}\mathbf{A} = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}$ has rank 1 too. $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ has rank 2, and $\mathbf{B}^\mathsf{T}\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has determinant $3 \neq 0$, so rank 2.

33.3. Row rank equals column rank

§12.4 stated that $\dim \operatorname{row}[\mathbf{A}] = \dim \operatorname{col}[\mathbf{A}]$ for every matrix and checked it for 2×2. Here is the proof.

Corollary 33.3. *For every* $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\operatorname{rank}(\mathbf{A}^\mathsf{T}) = \operatorname{rank}(\mathbf{A})$. *That is,* $\dim \operatorname{row}[\mathbf{A}] = \dim \operatorname{col}[\mathbf{A}]$.

Proof. Every column of $\mathbf{A}^\mathsf{T}\mathbf{A}$ is $\mathbf{A}^\mathsf{T}$ times a column of $\mathbf{A}$, hence a combination of the columns of $\mathbf{A}^\mathsf{T}$. So $\operatorname{col}[\mathbf{A}^\mathsf{T}\mathbf{A}] \subseteq \operatorname{col}[\mathbf{A}^\mathsf{T}]$, and a subspace inside another has no larger dimension (extend its basis by F2). Hence, with Corollary 33.2,

$$\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}^\mathsf{T}\mathbf{A}) \leq \operatorname{rank}(\mathbf{A}^\mathsf{T}).$$

This holds for *every* matrix, so it holds for $\mathbf{A}^\mathsf{T}$ in place of $\mathbf{A}$: $\operatorname{rank}(\mathbf{A}^\mathsf{T}) \leq \operatorname{rank}\big((\mathbf{A}^\mathsf{T})^\mathsf{T}\big) = \operatorname{rank}(\mathbf{A})$. The two inequalities together give equality. Finally $\operatorname{rank}(\mathbf{A}^\mathsf{T}) = \dim \operatorname{col}[\mathbf{A}^\mathsf{T}] = \dim \operatorname{row}[\mathbf{A}]$ (§11.1). □

33.4. When $\mathbf{A}^\mathsf{T}\mathbf{A}$ has full rank

Corollary 33.4. $\mathbf{A}^\mathsf{T}\mathbf{A}$ *has full rank* n *if and only if* $\mathbf{A}$ *has full column rank, i.e. the columns of* $\mathbf{A}$ *are independent.*

This is the hinge of least squares in Part 9: the system that produces the best approximate solution has the matrix $\mathbf{A}^\mathsf{T}\mathbf{A}$, and this corollary says when it has exactly one solution.

34. The 2×2 inverse

34.1. Undoing a matrix

The scalar equation $ax = b$ is solved by $x = a^{-1}b$ when $a \neq 0$, because $a^{-1}a = 1$. Part 1 §2.4 noted the resemblance between $ax = b$ and $\mathbf{Ax} = \mathbf{b}$ and warned that there was as yet no meaning for $\mathbf{A}^{-1}$. If there were a matrix $\mathbf{B}$ with $\mathbf{BA} = \mathbf{I}$, then multiplying $\mathbf{Ax} = \mathbf{b}$ by $\mathbf{B}$ would give $\mathbf{x} = \mathbf{BAx} = \mathbf{Bb}$.

Definition 34.1 (Inverse of a 2×2 matrix)**.** $\mathbf{A} \in \mathbb{R}^{2\times 2}$ is **invertible** if there is a matrix $\mathbf{B}$ with $\mathbf{AB} = \mathbf{BA} = \mathbf{I}$. Then $\mathbf{B}$ is the **inverse** of $\mathbf{A}$, written $\mathbf{A}^{-1}$.

There is at most one. If $\mathbf{BA} = \mathbf{I}$ and $\mathbf{AC} = \mathbf{I}$, then by associativity $\mathbf{B} = \mathbf{B}(\mathbf{AC}) = (\mathbf{BA})\mathbf{C} = \mathbf{C}$.

34.2. Finding it

Write $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and look for $\mathbf{X}$ with $\mathbf{AX} = \mathbf{I}$. Column by column (the column reading of a product), this is two systems: $\mathbf{A}\mathbf{x}_1 = \mathbf{e}_1$ and $\mathbf{A}\mathbf{x}_2 = \mathbf{e}_2$. The first is

$$ax + by = 1, \qquad cx + dy = 0.$$

Multiply the first equation by d, the second by b, and subtract: $(ad - bc)\,x = d$. Multiply the first by c, the second by a, and subtract: $(bc - ad)\,y = c$. So if $ad - bc \neq 0$,

$$\mathbf{x}_1 = \frac{1}{ad - bc} \begin{bmatrix} d \\ -c \end{bmatrix}, \qquad \text{and in the same way} \qquad \mathbf{x}_2 = \frac{1}{ad - bc} \begin{bmatrix} -b \\ a \end{bmatrix}.$$

The denominator is $\det \mathbf{A}$.

Theorem 34.1 (The 2×2 inverse)**.** *If* $\det \mathbf{A} = ad - bc \neq 0$*, then*

$$\boxed{\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.}$$

Proof. Multiply in both orders:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix}, \qquad \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix},$$

and divide by $ad - bc$. □

The recipe: *swap the diagonal entries, negate the off-diagonal entries, divide by the determinant.*

34.3. When there is no inverse

Theorem 34.2. *For* $\mathbf{A} \in \mathbb{R}^{2\times 2}$ *the following are equivalent:* $\mathbf{A}$ *is invertible;* $\det \mathbf{A} \neq 0$*;* $\operatorname{rank} \mathbf{A} = 2$*;* $\operatorname{null}[\mathbf{A}] = \{\mathbf{0}\}$.

Proof. If $\det \mathbf{A} \neq 0$, Theorem 34.1 produces the inverse. If $\mathbf{A}$ is invertible and $\mathbf{Ax} = \mathbf{0}$, then $\mathbf{x} = \mathbf{A}^{-1}\mathbf{Ax} = \mathbf{0}$, so the null space is $\{\mathbf{0}\}$. That gives rank 2 by rank–nullity, and rank 2 gives $\det \mathbf{A} \neq 0$ by §14.2. □

The geometry is the null space of §31.5. A matrix with $\det \mathbf{A} = 0$ merges different inputs into one output. No matrix $\mathbf{B}$ could send that one output back to two different places, so nothing can undo $\mathbf{A}$. A projection, which forgets a whole line, is the typical example.
One more property follows from P8 of §22: if $\mathbf{A}$ is invertible, $\det \mathbf{A} \det \mathbf{A}^{-1} = \det \mathbf{I} = 1$, so $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$. Undoing a matrix undoes its area factor.

34.4. The gallery, inverted

Example 34.1.

- **Rotation.** $\det \mathbf{R}_\theta = 1$, and the recipe gives $\mathbf{R}_\theta^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \mathbf{R}_{-\theta} = \mathbf{R}_\theta^{\mathsf{T}}$: rotate back (§24.6).

- **Shear.** $\begin{bmatrix}1 & k\\ 0 & 1\end{bmatrix}^{-1} = \begin{bmatrix}1 & -k\\ 0 & 1\end{bmatrix}$: slide back.
- **Scaling.** $\operatorname{diag}(s_1, s_2)^{-1} = \operatorname{diag}(1/s_1, 1/s_2)$, when $s_1 s_2 \neq 0$.
- **Reflection.** A reflection undoes itself: $\mathbf{F}^{-1} = \mathbf{F}$.
- **Orthogonal.** For any 2×2 orthogonal $\mathbf{Q}$, $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{Q}\mathbf{Q}^\mathsf{T} = \mathbf{I}$ says exactly $\mathbf{Q}^{-1} = \mathbf{Q}^\mathsf{T}$.
- **Projection.** $\det \mathbf{P_u} = 0$: no inverse.

34.5. Solving with it

Part 1 §1.5 solved $\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}\mathbf{x} = \begin{bmatrix}5\\ 5\end{bmatrix}$ by substitution and got $\mathbf{x} = [\,2, 1\,]$. Now $\det = 6 - 1 = 5$ and

$$\mathbf{x} = \mathbf{V}^{-1}\mathbf{u} = \frac{1}{5}\begin{bmatrix}3 & -1\\ -1 & 2\end{bmatrix}\begin{bmatrix}5\\ 5\end{bmatrix} = \frac{1}{5}\begin{bmatrix}10\\ 5\end{bmatrix} = \begin{bmatrix}2\\ 1\end{bmatrix}. \quad \checkmark$$

This is the matrix meaning of $x = b/a$ promised in Part 1 §2.4.

Inverses of products reverse the order, like transposes (M6): if $\mathbf{A}$ and $\mathbf{B}$ are invertible then $(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}(\mathbf{B}\mathbf{B}^{-1})\mathbf{A}^{-1} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$, so $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$. To undo "$\mathbf{B}$, then $\mathbf{A}$", undo $\mathbf{A}$ first.

34.6. Practice

P1. Find the null space and nullity of $\begin{bmatrix}1 & -1 & 2\\ 2 & -2 & 4\end{bmatrix}$, and check rank–nullity.

P2. Without computing, give the smallest possible nullity of a 5×8 matrix and the largest possible rank of a 7×3 matrix.

P3. Find all solutions of $x_1 + 2x_2 - x_3 = 4$ (one equation, three unknowns) in the form $\mathbf{x}_p + \text{null}[\mathbf{A}]$.

P4. Which lines through the origin are invariant under $\operatorname{diag}(2, 3)$? Under $3\mathbf{I}$?

P5. For $\mathbf{A} = \begin{bmatrix}1 & 1\\ 1 & 1\\ 0 & 0\end{bmatrix}$, compute $\mathbf{A}^\mathsf{T}\mathbf{A}$ and check that the ranks agree.

P6. Invert $\begin{bmatrix}4 & 7\\ 2 & 6\end{bmatrix}$ and use the inverse to solve $\begin{bmatrix}4 & 7\\ 2 & 6\end{bmatrix}\mathbf{x} = \begin{bmatrix}1\\ 2\end{bmatrix}$.

Answers

P1. Both rows say $x_1 - x_2 + 2x_3 = 0$. With $x_2 = s$ and $x_3 = t$ free, $x_1 = s - 2t$, so $\text{null}[\mathbf{A}] = \text{span}\{[\,1, 1, 0\,], [\,-2, 0, 1\,]\}$, nullity 2. The rank is 1, and $1 + 2 = 3$. ✓

P2. Rank at most 5, so nullity at least $8 - 5 = 3$. Rank at most $\min(7, 3) = 3$.

P3. $\mathbf{x}_p = [\,4, 0, 0\,]$; the null space of $\begin{bmatrix}1 & 2 & -1\end{bmatrix}$ is $\text{span}\{[\,-2, 1, 0\,], [\,1, 0, 1\,]\}$; so $\mathbf{x} = [\,4, 0, 0\,] + s[\,-2, 1, 0\,] + t[\,1, 0, 1\,]$, a plane in $\mathbb{R}^3$ not through the origin.

P4. $\operatorname{diag}(2, 3)$: only the two axes. If $[\,x, y\,] \mapsto [\,2x, 3y\,]$ is a multiple $\lambda[\,x, y\,]$ with x, y both nonzero, then $\lambda = 2 = 3$, impossible. $3\mathbf{I}$: every line, since every vector is only stretched.

P5. $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, rank 1; $\mathbf{A}$ has two equal columns, rank 1. ✓

P6. $\det = 24 - 14 = 10$, so $\mathbf{A}^{-1} = \frac{1}{10}\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$ and $\mathbf{x} = \frac{1}{10}[\,6-14,\ -2+8\,] = [\,-0.8,\ 0.6\,]$. Check: $4(-0.8) + 7(0.6) = 1$ and $2(-0.8) + 6(0.6) = 2$. ✓

Looking ahead. The 2×2 determinant has done a great deal: it measures area, detects rank, and builds the inverse. Part 7 defines the determinant of an $n \times n$ matrix, by expanding along a row, and proves the properties that make it as useful in every dimension as it has been in the plane.